

Exam Mastery Guide: Social and Professional Issues in Computing

1. Strategic Exam Overview & Priority Matrix

To achieve an A+ in the Daffodil International University (DIU) CSE431/CSE498 curriculum, you must move beyond passive reading of Sara Baase's *A Gift of Fire*. Success is determined by your ability to weaponize theoretical frameworks against complex technical scenarios. Securing the 10 marks typically allocated to Section 1 (CO2/CO3) is the single most important factor for an elite grade. Analysis of recent DIU papers (Fall 2025, Spring 2026, Spring 2023) reveals that examiners do not just want definitions; they want you to act as a Senior Lead who can balance stakeholder rights against professional liability.

Topic Priority Matrix

Topic, Priority, Reasoning

Professional Ethics Scenarios, HIGH, Mandatory for Section 1; carries 25% of total exam marks. Focus on CO2/CO3.

Intellectual Property (Copyright/Patent), HIGH, Guaranteed in CO4. Must cite Bangladesh-specific Acts (2022/2023).

System Reliability & Failures, HIGH, Critical for CO3/CO5. Focus on automated deployment and testing failures.

The Digital Divide & Evaluating Info, MEDIUM, Usually appears as a 5-mark conceptual question in CO1.

Neo-Luddite Views, LOW, Primarily used for critical discussion or 2.5-5 mark sub-questions.

Common Question Rewording Patterns

The "Scenario Template" is the examiner's favorite tool. Whether the character is **Mahi** (Hospital), **Rumi** (Fire Systems), **Nerob** (Smart Gadgets), or **Jahid** (PDF sharing), they are all testing your ability to identify the same core conflict: the tension between institutional pressure (deadlines/profit) and professional duty (safety/integrity). If the scenario mentions a "smart" device (like Nerob's kitchen gadgets) or an "app" (like Spring 2025), you are being tested on a mix of IP law and Professional Ethics. *The following sections provide the rigorous methodology required to dismantle these questions.*

2. Domain 1: Professional Ethics and Codes of Conduct (PRIORITY: HIGH)

Professional ethics is the anchor of the exam. You are expected to demonstrate that a computer professional has a higher duty than a layperson because their work affects life-critical systems.

Core Responsibilities (Sara Baase Framework)

According to *A Gift of Fire*, your answers must reflect these core professional duties:

- **Understand Success:** Success is not just a compiled code; it is a system that meets user needs safely.
- **Include Users:** Failure to include medical staff, pilots, or end-users in testing is a breach of professional conduct.

- **Honesty about Capabilities:** You must be transparent about what software *cannot* do, especially regarding its safety limitations.
- **Don't Assume Safety:** Never assume a legacy system or an automated update is correct; it must be reviewed and tested.

Methodology for Analyzing Scenarios

You must apply this two-phase methodology to every scenario to secure maximum marks:

- **Brainstorming Phase:**
- **Stakeholders:** *Mandatory.* List every party affected (e.g., Users, Employers, Public, Regulators).
- **Risks/Consequences:** Identify harms (financial, physical, or legal).
- **Action Options:** Provide alternatives to a simple "Yes/No" choice.
- **Analysis Phase:**
- **Normative Ethics Application:** Identify the framework. Are you acting out of **Deontology** (Duty-based: "I must not lie because it is my duty") or **Utilitarianism** (Consequence-based: "I will hide the bug to prevent a 1% chance of failure and save the company money")?
- **Categorization:** Label the action as **Ethically Obligatory** , **Prohibited** , or **Acceptable** .

Model Answer Guide: The "Mahi" and "Rumi" Scenarios

- **Scenario 1: Mahi (Hospital Database):** When Mahi refuses to lie about a bug to generate insurance revenue, he is demonstrating **Deontological Ethics** (Duty to Honesty).
- **Stakeholders:** Mahi, the Manager, Hospital Board, Patients, Insurance Companies, and Government Regulators.
- **Justification:** Mahi prioritizes professional integrity and legal compliance over the manager's utilitarian goal of short-term revenue.
- **Scenario 2: Rumi (Fire Control Systems):** Rumi initially follows **Professional Responsibility** by reporting a flaw. However, her later decision to continue based on "financial loss" concerns represents a shift toward a flawed **Utilitarian** perspective.
- **Stakeholders:** Rumi, the Project Manager, Higher Authority, Firefighters (end-users), and potential fire victims.
- **Ethics Type:** Her initial refusal to skip reporting is **Normative/Deontological** (Professional Duty).

Code of Ethics: Roles and Limitations

Positive Roles, Limitations

"Provides a ""Moral Victory"" and acknowledging the issue.", Cannot provide a magic formula for every controversy.

Establishes high standards of conduct and integrity., Can be ignored by institutionalized bureaucracies.

Educates professionals about their obligations., Some ethical issues remain inherently controversial.

3. Domain 2: Intellectual Property – Laws and Scenarios (PRIORITY: HIGH)

IP rights are territorial and essential for protecting tech innovation. In the Bangladesh context, you must cite the most recent legislation.

Copyright vs. Patent: The Bangladesh Standard

Feature, Copyright, Patent

Protects, Creative/Literary works and Software., "Inventions (New, useful, non-obvious)."

BD Legislation, Copyright Act 2023, Patents Act 2022

Duration (BD), Life + 60 years, 20 years from filing

Registration, Often automatic upon creation., Strictly required.

Model Case: Nerob's Smart Kitchen Gadgets

In the scenario where **Nerob** develops a "Music Air Fryer" with a unique logo and trending songs:

- **IP Identification:** The unique logo is a **Trademark**. The energy-saving fryer technology is a **Patent**. The "trending songs" are protected by **Copyright**.
- **Violation:** Using famous songs without a license violates professional responsibility toward copyright owners.
- **Solutions for IP Violations (The Three Groups):**
- **Technical:** Use Digital Watermarks or Content ID systems to track unauthorized re-uploads (as in the **Raju** scenario).
- **Legal:** File for protection under the **Copyright Act 2023** or **Patents Act 2022**; pursue cease-and-desist orders.
- **Social/Ethical:** Promote "Fair Dealing" awareness and ensure proper licensing agreements are in place before distribution (as in the **Jahid** PDF scenario).

Territorial Nature & OTT Platforms

IP is **territorial**. Platforms like **Netflix** must obtain specific licenses that dictate the **duration** and **territory** for every movie. A license in the US does not automatically grant rights in Bangladesh.

4. Domain 3: Computer Errors, Failures, and Reliability (PRIORITY: HIGH)

Categories of Failure & Real-Life Evidence

1. **Programming Bugs:** Small errors with massive impacts (e.g., the insurance bug in the Mahi case).
2. **Data Entry/System Errors:** The **\$15 million bank error** where an employee forgot a command, affecting 150,000 accounts.
3. **Operational Failures:** The **Gazipur Garments Factory** case—failure caused by using an **old system** with an **untested update**.

Case Study: Cloud Infrastructure Outage (Spring 2026)

- **Type of Failure:** Sudden global operational outage.

- **Key Causes:** Inadequate testing , excessive reliance on automated deployment , and the **absence of effective fallback mechanisms** .
- **Stakeholders:** The Infrastructure Provider, millions of users, businesses, and financial systems.
- **Management Steps for Safety:**
- Require a **convincing case for safety** before any deployment.
- Never assume software is safe just because "it usually works."
- Implement **manual overrides** for automated deployment systems.

5. Domain 4: Social Issues – Digital Divide & Agriculture (PRIORITY: MEDIUM)

The Digital Divide & Wisdom of the Crowd

- **The Divide:** Shifting from "Haves" vs. "Have-nots" to **"Haves" vs. "Have-Laters."** Technology prices (like disk storage at \$100/TB) eventually lower the barrier to entry.
- **Wisdom of the Crowd:** This phenomenon only works when the crowd is **diverse and independent** . If there is peer pressure or dominant personalities (as noted in *A Gift of Fire*), the "wisdom" fails.

Agriculture Automation: The Five Arguments (Spring 2023)

If asked whether automation is the best solution for agriculture, provide these arguments:

1. **Labor Shortage:** Declining appeal of manual labor due to urbanization and aging populations.
2. **Efficiency:** Automated tech improves crop yields over longer working hours.
3. **Reliability:** Robots provide more consistent results than manual labor in harsh conditions.
4. **Cost Barrier:** It is prohibitively expensive for low-income or mid-level income farmers.
5. **Sustainability vs. Production:** High production may come at the cost of failing to provide a cost-effective plan for small-scale farmers.

6. Last-Minute High-Yield Revision Section

Term/Definition Blitz

- **Professional Ethics:** Responsibilities toward customers, clients, and the public whom your products affect.
- **Stakeholders:** *Mandatory Answer Component.* All people/orgs affected by a system.
- **Fair Dealing:** An exception to copyright law (e.g., for education), but it does not allow sharing an entire scanned textbook (Jahid's case).
- **Greenhouse Effect:** In complex modeling, the phenomenon where gases like CO2 trap heat; models are simplifications and contain "parametrizations" for unknown science.
- **Neo-Luddites:** Critics who view technology as "Pandora's Box," citing de-skilling of jobs and social disintegration.

Bangladesh Law & Status

1. **Copyright Act 2023:** Protects software and creative works.

2. **Patents Act 2022:** Protects non-obvious inventions.
3. **Cryptocurrency:** Viewed as a **Computer Crime** in Bangladesh (Spring 2023 Paper) due to unauthorized digital asset exchange and potential for money laundering. **FINAL EXAM TIP:** To secure the highest marks in scenario questions, **always list the Stakeholders first.** Whether the case is a "Phishing Scam" or a "Cloud Outage," identifying the human impact is the key to demonstrating mastery of CO2 and CO3.